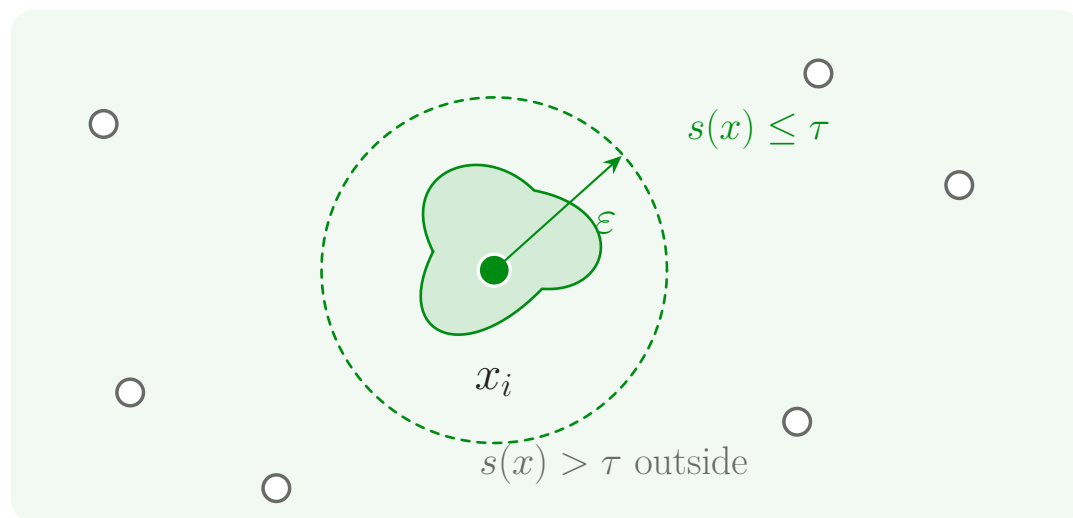


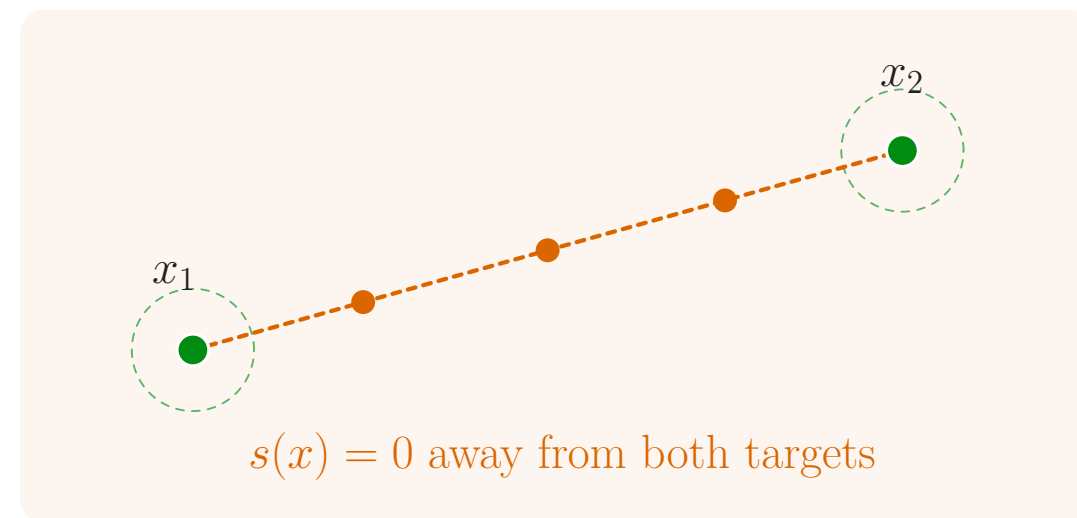
When can a low score be trusted?

$$s(x) = \frac{\|C\phi(x)\|}{\|A_T\phi(x)\|}, \quad \mathcal{X} = g(\mathcal{Z})$$

The desired separation



Affine-input counterexample



ϕ affine; admissible blends with nonzero denominator

$$\gamma_{\mathcal{X}}(\epsilon) = \inf_{\substack{x \in \mathcal{X} \\ \min_i \|x - x_i\| > \epsilon}} s(x)$$

$$s(x) \leq \tau < \gamma_{\mathcal{X}}(\epsilon) \implies \min_i \|x - x_i\| \leq \epsilon$$

$x \in \mathcal{X}, \quad \|A_T\phi(x)\| > 0$ target: a positive wrong-image score floor on the allowed family